Quiz #6

1. For a confidence interval for a population mean, determine how the **width** changes in each of the following situations, assuming all other quantities remain constant.

   (a) The sample size n increases.

   (b) The confidence level increases.

   (c) The population mean μ increases.

   (d) The population standard deviation σ increases.

   For each part, choose one of the following answers:

   - Gets wider
   - Gets narrower
   - Stays the same

   (a) **Gets narrower.** Increasing n reduces the standard error.

   (b) **Gets wider.** A higher confidence level requires a larger critical value.

   (c) **Stays the same.** The population mean changes the center, not the width.

   (d) **Gets wider.** Increasing σ increases the standard error.

2. Fill in the blanks using **null** or **alternative**.

   The _____ hypothesis says that an observed difference can be explained by chance variation, while the _____ hypothesis proposes that the difference reflects a real difference in the population.

   Null; Alternative.

   The null hypothesis attributes the observed difference to chance variation, while the alternative hypothesis proposes a real difference in the population.

3. Consider the following P-values:

$$0.1\%, \qquad 3\%, \qquad 17\%, \qquad 32\%.$$

   (a) Which P-value provides the strongest evidence for retaining the null hypothesis? Explain briefly.

   (b) Which P-value provides the strongest evidence against the null hypothesis? Explain briefly.

(a) 32%. A larger P-value indicates weaker evidence against the null hypothesis. However, it does not prove that the null hypothesis is true.

(b) 0.1%. A smaller P-value indicates stronger evidence against the null hypothesis. The observed result is unlikely under the null hypothesis.

4. State whether each statement is **true** or **false** and briefly explain.

   (a) The observed significance level depends on the data.

   (b) If the observed significance level is 5%, there is a 95% probability that the alternative hypothesis is true.

(a) **True.** The observed significance level (P-value) is calculated from the data and depends on the observed test statistic.

(b) **False.** A P-value of 5% means that, assuming the null hypothesis is true, there is a 5% probability of obtaining a test statistic as extreme as or more extreme than the observed one. It does not give the probability that either hypothesis is true.

5. A NumPy array named `readings` contains independent measurements from an approximately normal population. The population SD is unknown.

```
from scipy import stats

n = len(readings)

xbar = readings.mean()

se = stats.sem(readings)

t_star = stats.t.ppf(0.975, df=n-1)

ci = (xbar - t_star*se, xbar + t_star*se)
```

   (a) In words, explain what the code calculates. What are `se`, `t_star`, and `ci`?

   The code constructs a **95% Student's t confidence interval for the population mean**. The variable `se` is the estimated standard error of the sample mean, $s/\sqrt{n}$. The variable `t_star` is the t critical value with $n-1$ degrees of freedom. The variable `ci` is a pair containing the lower and upper confidence bounds.

   (b) Why does the code use `0.975` in `stats.t.ppf` for a 95% confidence interval?

   A two-sided 95% interval leaves 5% outside its bounds, or 2.5% in each tail. The positive critical value is therefore the point with 97.5% of the t distribution to its left: $1 - 0.025 = 0.975$.